

ASSIGNMENT

Design and Develop a Database for a Smart Campus Management System

Objective

Design a complete database for a Smart Campus Management System by applying database design principles and implementing SQL queries.

Modules (Choose any 5)

- Student Management
- Faculty Management
- Attendance
- Library
- Hostel
- Examination
- Fee Management
- Club Activities
- Sports
- Placement Cell

Students must complete the following tasks:

Part A – Requirement Analysis

- Objectives
- Users of the system
- Functional requirements
- Non-functional requirements

Part B – Database Design

- ER Diagram
- Relational Schema

- Identify Primary Keys and Foreign Keys
- Normalize all tables up to **Third Normal Form (3NF)**

Part C – Database Implementation

- Create **minimum 4 tables**
- Use appropriate constraints:
 - PRIMARY KEY
 - FOREIGN KEY
 - UNIQUE
 - NOT NULL
 - CHECK
 - DEFAULT

Part D – Data Entry

Insert **minimum 20 records** into each main table.

Part E – SQL Queries

Write:

- 10 Retrieval Queries
- 5 Aggregate Function Queries
- 5 JOIN Queries
- 5 Nested Queries
- 5 UPDATE Queries
- 5 DELETE Queries
- 5 ALTER TABLE Queries
- 5 Pattern Matching Queries (LIKE)

Part F – Report

Include:

- Screenshots

- SQL Code
- Output
- Challenges faced
- Learning outcomes

Deliverables

- SQL File
- 12–20 page Report
- 5-minute Presentation (PPT for reference)

Note: Last date of assignment submission is 27th August,2026. No assignment will be entertained after the given deadline. Also note that you have to give your assignment in pink cover page tied with a blue ribbon. The pink cover page will include your details such as name, roll number, class etc.